

PHASE 4 – Feature-wise Implementation Plan

Project: Prism IQ

Version: 1.0

Status: Development Roadmap

Development Methodology: Feature-Driven Development (FDD)

Sprint Strategy: MVP First

Development Philosophy

Every feature must satisfy four conditions:

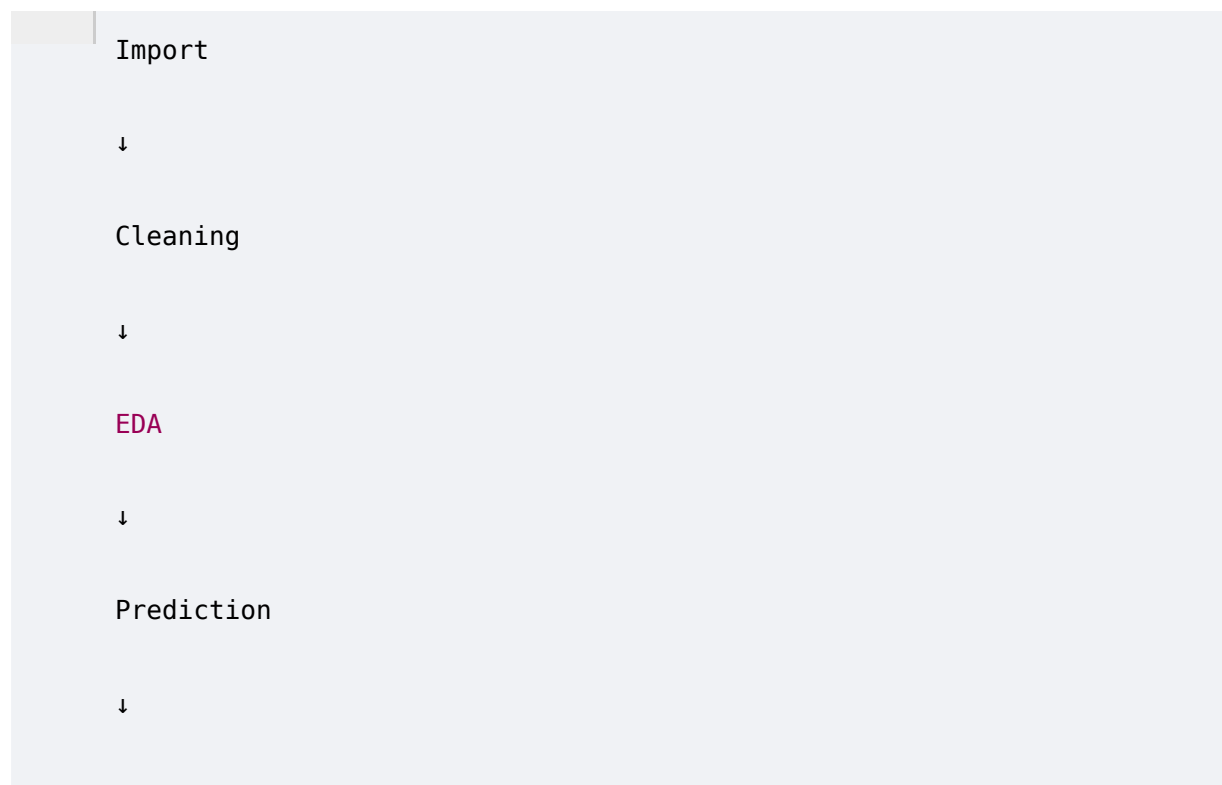
-  Independent
-  Testable
-  Modular
-  Demo-ready

Each feature should work even if later features are incomplete.

Project Milestones

Instead of building frontend first or backend first,

Prism IQ will be developed **Pipeline First**.



Every milestone completes one stage of the analytics pipeline.



Milestone 1 — Project Foundation

Goal

Prepare the entire development environment.

Frontend

- Initialize React + Vite
 - Configure TypeScript
 - Install Tailwind CSS
 - Configure shadcn/ui
 - Install Recharts
 - Install Plotly
 - Setup Routing
 - Setup Layout
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Backend

- Initialize FastAPI
 - Configure SQLAlchemy
 - Configure PostgreSQL
 - Setup Alembic
 - Environment Variables
 - Logging
 - Folder Structure
-

Deliverable

Project runs successfully on localhost.

Estimated Time

4–5 hours



Milestone 2 — Data Import Module

Goal

Support importing structured datasets.

Features

CSV Upload

Excel Upload

API Import

HTML Table Extraction

Backend Tasks

Create upload endpoints.

Validate files.

Save raw dataset.

Create Dataset metadata.

Generate Dataset ID.

Frontend Tasks

Upload page.

Drag & Drop.

Import progress.

Success screen.

Acceptance Criteria

User can import supported data sources.



Milestone 3 — Data Profiling

Goal

Automatically understand uploaded data.

Generate

Dataset Shape

Data Types

Memory Usage

Missing Values

Duplicate Rows

Unique Values

Statistics

Column Summary

Backend

Pandas Profiling Engine

Metadata Storage

Quality Score

Frontend

Overview Cards

Quality Indicators

Column Table

Acceptance Criteria

Automatic profile generated after upload.



Milestone 4 — Smart Cleaning

Goal

Clean datasets automatically.

Cleaning Features

Missing Values

Duplicates

Invalid Types

Categorical Standardization

Optional Outlier Handling

Backend

Cleaning Service

Cleaning History

Cleaned Dataset Storage

Frontend

Cleaning Preview

Cleaning Summary

Download Cleaned Dataset

Acceptance Criteria

User receives cleaned dataset.



Milestone 5 — Exploratory Data Analysis (EDA)

Goal

Generate intelligent visualizations.

Charts

Histogram

Bar

Pie

Scatter

Line

Box Plot

Heatmap

Correlation Matrix

Backend

EDA Engine

Chart Data API

Correlation Service

Frontend

Interactive Dashboard

Chart Controls

Filtering

Acceptance Criteria

Charts generated automatically.



Milestone 6 — Predictive Analytics

Goal

Train ML models.

Workflow

User selects target



Problem Detection



Train Multiple Models



Compare Metrics



Select Best



Prediction

Models

Regression

Linear Regression

Decision Tree

Random Forest

Classification

Logistic Regression

Decision Tree

Random Forest

Frontend

Target Selector

Training Progress

Model Comparison

Prediction Results

Acceptance Criteria

Best model selected automatically.



Milestone 7 — Recommendation Engine

Goal

Generate business intelligence.

Generate

Business Insights

Patterns

Anomalies

Recommendations

Priority

Confidence Score

Backend

Recommendation Generator

Insight Engine

Rule Engine

Frontend

Insight Cards

Priority Cards

Business Summary

Acceptance Criteria

Recommendations generated automatically.



Milestone 8 — Dashboard

Goal

Single analytics workspace.

Dashboard Pages

Home

Dataset

Cleaning

EDA

Prediction

Insights

Reports

Widgets

KPIs

Charts

Model Performance

Recommendations

Downloads

Acceptance Criteria

Entire workflow visible.



Milestone 9 — Report Generation

Goal

Generate executive-ready reports.

Reports

Executive Report

Business Report

Cleaned Dataset

PDF Sections

Executive Summary

Cleaning Summary

EDA Highlights

Prediction Results

Recommendations

Final Health Score

Acceptance Criteria

Professional PDF generated.



Milestone 10 — Final Testing

- Testing
- Functional
- API
- UI
- Integration
- Performance
- Edge Cases

- Deliverables
- Working Project
- Documentation
- GitHub Repository
- Deployment
- Presentation

Feature Dependency Graph

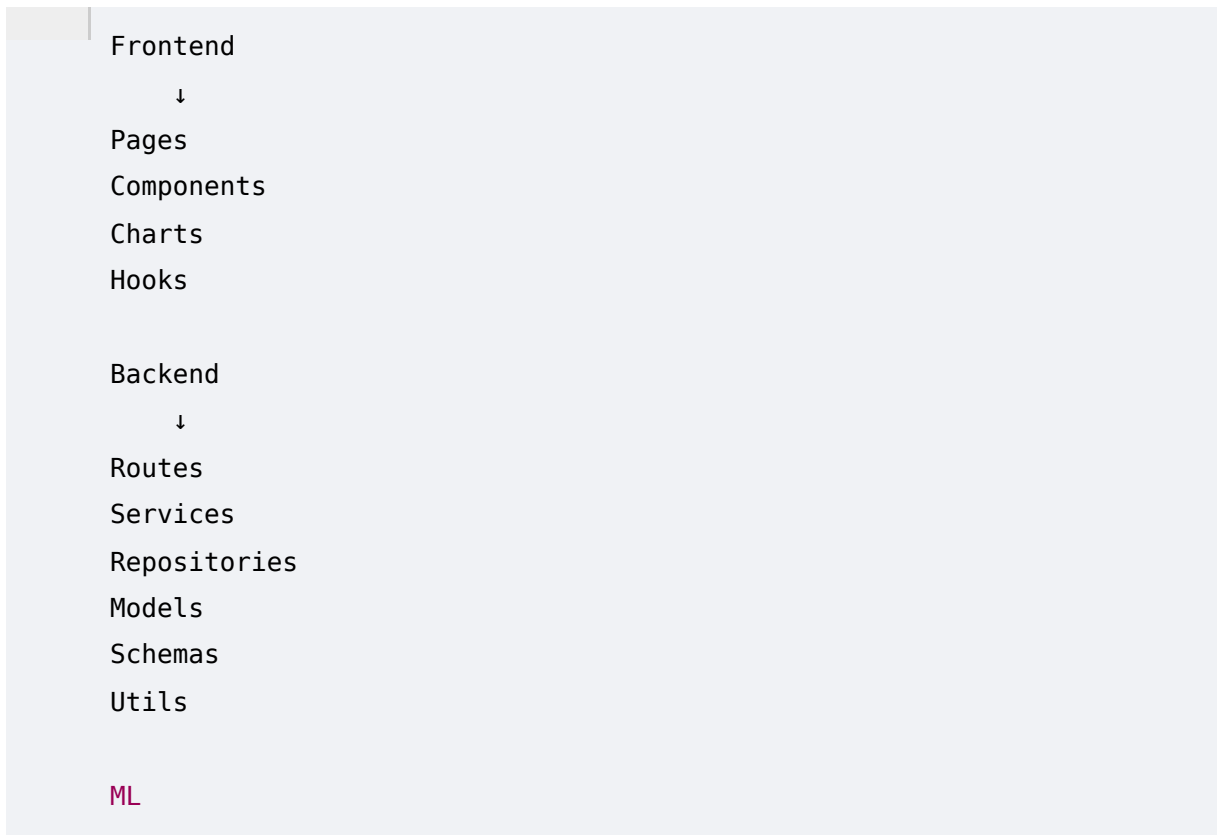


Feature Priority Matrix

Priority	Feature	Status
 Critical	Project Foundation	Must Build
 Critical	Data Import	Must Build
 Critical	Data Profiling	Must Build
 Critical	Smart Cleaning	Must Build
 Critical	EDA	Must Build
 Critical	Predictive Analytics	Must Build
 Critical	Dashboard	Must Build
 Critical	Report Generation	Must Build
 High	Recommendation Engine	Strongly Recommended
 Nice to Have	API History	Future
 Nice to Have	Project History	Future

● Nice to Have	User Accounts	Future
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Folder Ownership (Logical)



Development Rules

Every feature must satisfy:

- API implemented first.
- Backend tested before frontend integration.
- Every endpoint documented.
- Every feature demoable independently.
- No placeholder implementations.
- No duplicate business logic.
- Preserve the original uploaded dataset; all cleaning outputs are written as new artifacts.

MVP Completion Checklist

Data Pipeline

- CSV Upload
- Excel Upload
- API Import
- HTML Table Extraction

Analytics

- Profiling
- Cleaning
- EDA
- Dashboard

Machine Learning

- Target Selection
- Multi-model Training
- Best Model Selection
- Prediction

Intelligence

- Recommendations
- Business Insights

Reports

- Executive PDF
 - Cleaned CSV
-

Architecture Review

Veere, eh roadmap strong aa, par main ik improvement recommend karaanga.

Current milestones are **feature-based**, but Prism IQ di nature **pipeline-based** aa. So implementation de time assi har milestone nu internally **vertical slices** ch todange.

Example:

Vertical Slice 1

- Upload API
- Upload UI
- Upload DB
- Upload Testing

Done. 

Fer next:

Vertical Slice 2

- Profiling API
- Profiling UI
- Profiling DB
- Profiling Testing

Done. 

Is approach de benefits:

- Har 1–2 din baad working demo ready.
- Bugs early stage te hi mil jaan ge.
- Frontend te backend parallel synchronize rahange.
- Friday deadline de hisaab naal risk kaafi ghatt ho ju.

Mere hisaab naal eh Prism IQ layi sab ton practical development strategy hovegi. 